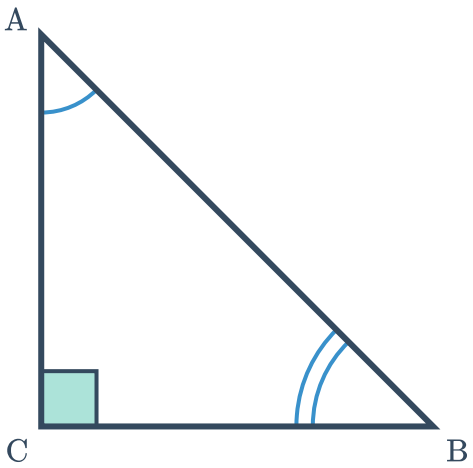
Hypotenuse

1 For each of the following, find the number that completes a Pythagorean triple given the two smallest numbers:

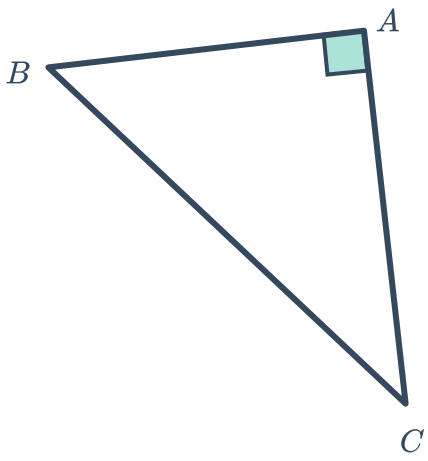
a 3 and 4 **b** 8 and 6 **c** 9 and 12 **d** 8 and 15

2 For each diagram, identify which side of the triangle is the hypotenuse.

a



b



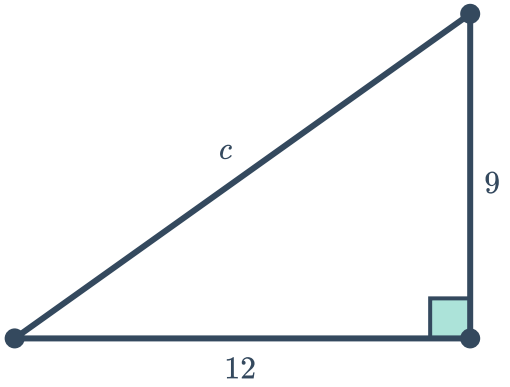
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3 For the following right-angled triangles:

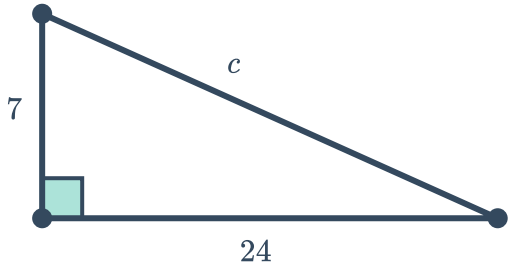
i Write the equation that the sides of the triangle satisfy.

ii Solve the equation to find the length of the hypotenuse.

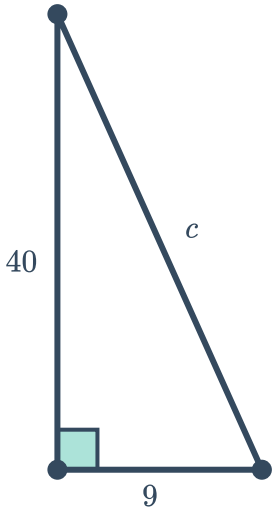
a



b



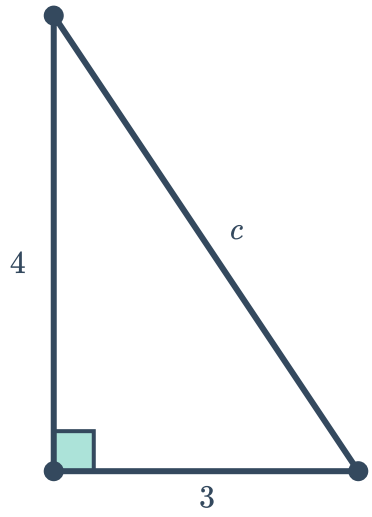
c



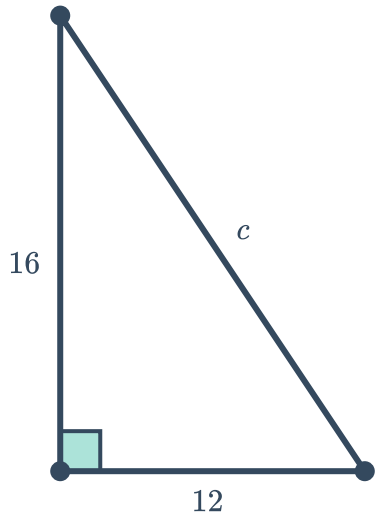
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4 Find the length of the unknown side of the following triangles:

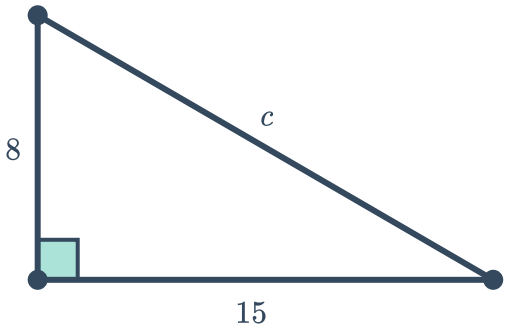
a



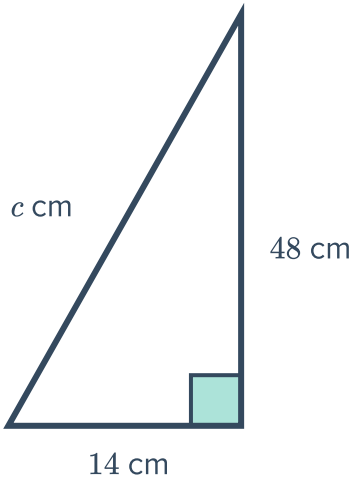
b



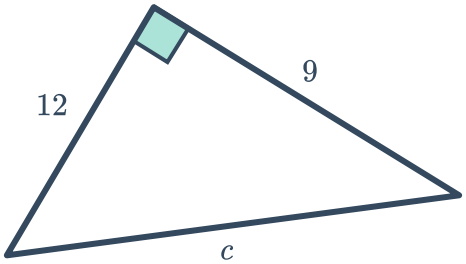
c



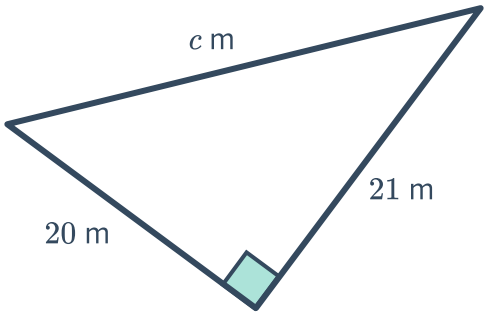
d



e



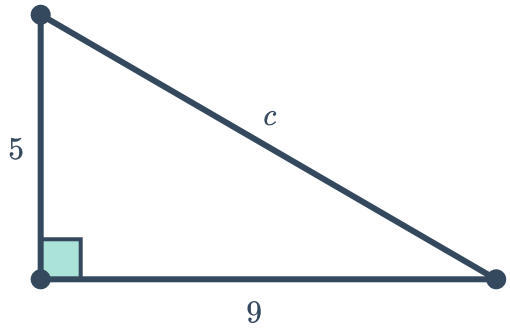
f



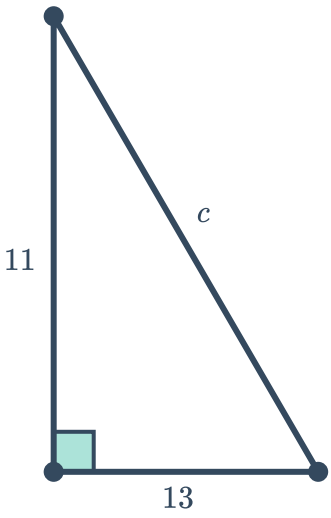
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5 Find the length of the unknown side of the following triangles as a surd:

a

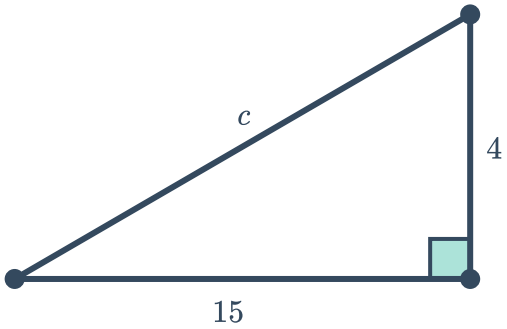


b

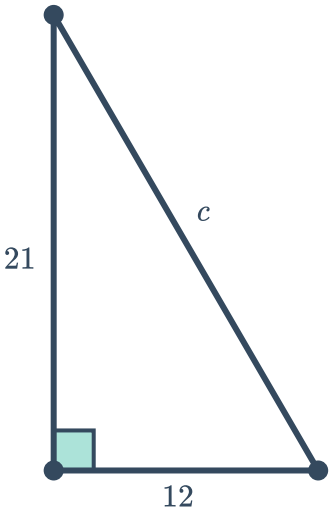


6 Find the length of the unknown side of the following triangles, correct to two decimal places:

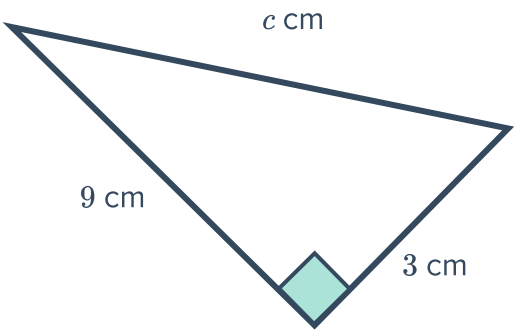
a



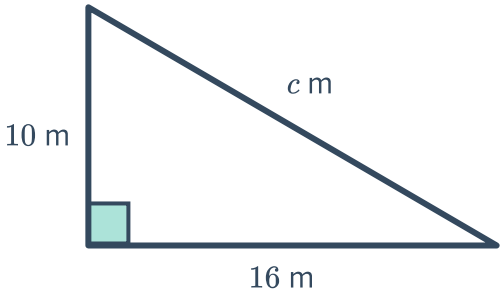
b



c

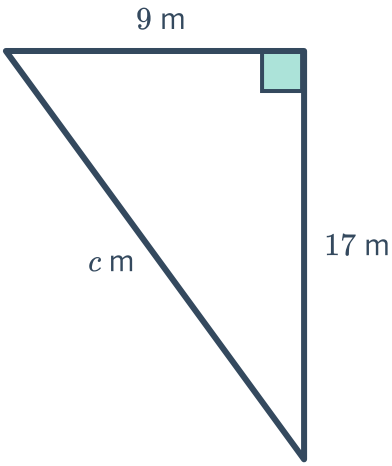


d

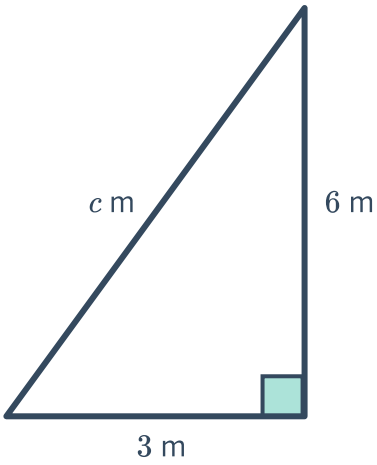


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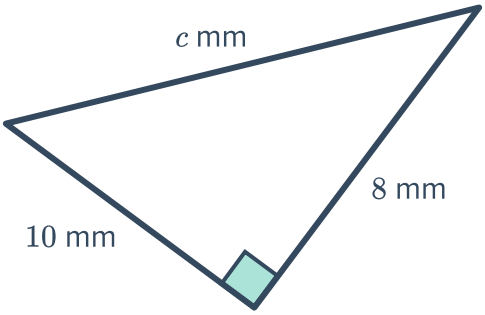
e



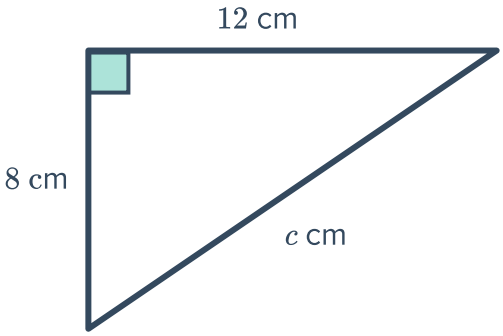
f



g



h

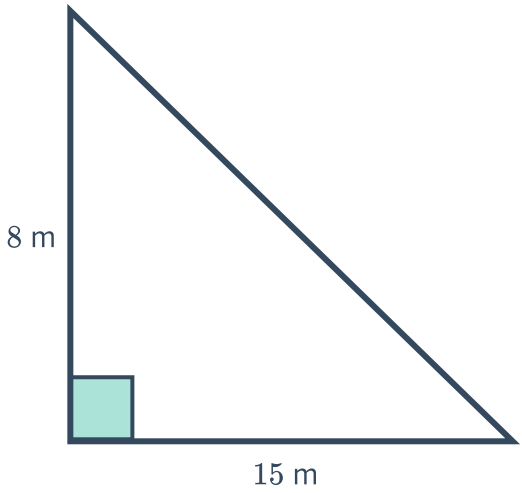


Perimeter

7 Consider the following triangle:

a Find the length of the hypotenuse.

b Find the perimeter of the triangle.



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a Find the length of the hypotenuse, correct to two decimal places.

b Find the perimeter of the triangle, correct to two decimal places.

9 A right isosceles triangle has two sides measuring $17\,\text{cm}$.

a Find the length of its hypotenuse, correct to two decimal places.

b Find the perimeter of the triangle, correct to two decimal places.

Diagonals of rectangles

10 Find the exact length of the diagonals of the following rectangles:

a A rectangle whose side lengths are 9 and 12.

b A rectangle whose side lengths are 9 and 14.

11 Find the length of the diagonals of the following squares, correct to two decimal places:

a A square with side length of 9.

b A square with side length of 12.

12 Complete the following table:

Side length of square	Exact length of diagonal
1	
3	
6	
11	
249	

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